Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · VAR

several interrelated series

Table 1. Lag selection

Lag	AIC	BIC	HQ
1	−8.25	−7.96	−8.13
2	−8.25	−7.73	−8.04
3	−8.23	−7.50	−7.93
4	−8.20	−7.24	−7.81
5	−8.30	−7.12	−7.82
6	−8.22	−6.82	−7.65
7	−8.21	−6.59	−7.55
8	−8.14	−6.30	−7.39
9	−8.10	−6.04	−7.27
10	−8.13	−5.85	−7.21

Table 2. VAR coefficients (per equation)

	gdp_growth	inflation	unemployment
gdp_growth.l1	0.44	0.22	−0.82
	(0.11)	(0.07)	(0.05)
	[0.22, 0.67]	[0.07, 0.37]	[−0.91, −0.72]
inflation.l1	0.18	0.09	0.16
	(0.18)	(0.12)	(0.08)
	[−0.18, 0.54]	[−0.15, 0.32]	[0.00, 0.32]
unemployment.l1	−0.34	−0.15	−0.14
	(0.23)	(0.15)	(0.10)
	[−0.80, 0.13]	[−0.45, 0.15]	[−0.34, 0.06]
gdp_growth.l2	0.26	−0.05	−0.11
	(0.23)	(0.15)	(0.10)
	[−0.19, 0.71]	[−0.35, 0.24]	[−0.31, 0.08]
inflation.l2	0.00	0.19	−0.02
	(0.19)	(0.12)	(0.08)
	[−0.37, 0.37]	[−0.05, 0.43]	[−0.18, 0.14]
unemployment.l2	−0.11	0.10	−0.12
	(0.23)	(0.15)	(0.10)
	[−0.57, 0.36]	[−0.20, 0.40]	[−0.32, 0.08]
gdp_growth.l3	0.11	0.07	−0.12
	(0.23)	(0.15)	(0.10)
	[−0.35, 0.56]	[−0.23, 0.37]	[−0.31, 0.08]
inflation.l3	−0.12	0.12	0.12
	(0.18)	(0.12)	(0.08)
	[−0.48, 0.24]	[−0.11, 0.36]	[−0.03, 0.28]
unemployment.l3	−0.12	−0.21	−0.19
	(0.23)	(0.15)	(0.10)
	[−0.59, 0.34]	[−0.51, 0.09]	[−0.39, 0.01]
gdp_growth.l4	−0.00	−0.13	−0.14
	(0.23)	(0.15)	(0.10)
	[−0.45, 0.45]	[−0.43, 0.16]	[−0.34, 0.06]
inflation.l4	−0.06	−0.05	0.02
	(0.18)	(0.12)	(0.08)
	[−0.43, 0.30]	[−0.29, 0.19]	[−0.14, 0.17]
unemployment.l4	0.11	−0.21	0.02
	(0.24)	(0.15)	(0.10)
	[−0.36, 0.58]	[−0.51, 0.10]	[−0.19, 0.22]
gdp_growth.l5	−0.09	−0.22	0.03
	(0.22)	(0.14)	(0.10)
	[−0.53, 0.35]	[−0.50, 0.07]	[−0.16, 0.22]
inflation.l5	0.01	−0.11	0.06
	(0.18)	(0.12)	(0.08)
	[−0.35, 0.37]	[−0.35, 0.12]	[−0.10, 0.22]

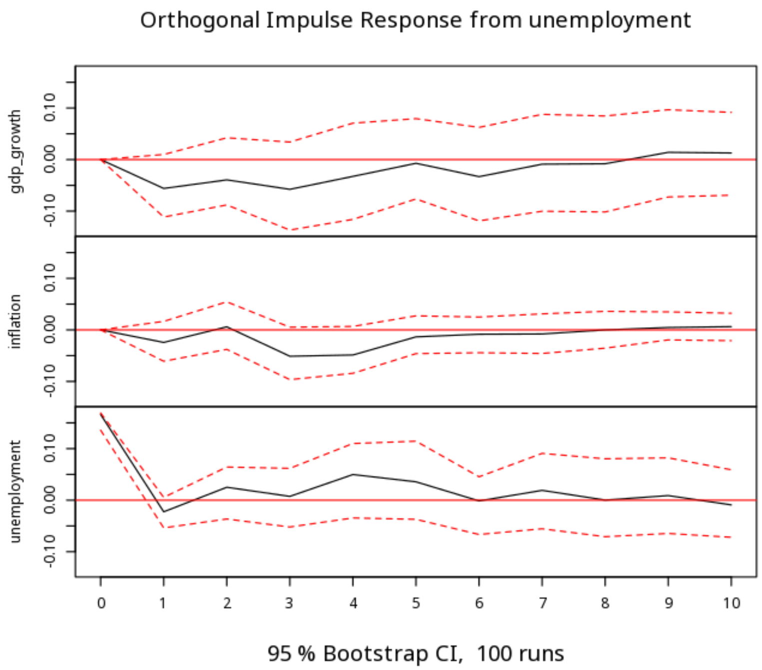
	gdp_growth	inflation	unemployment
unemployment.l5	0.35	0.03	0.06
	(0.11)	(0.07)	(0.05)
	[0.13, 0.56]	[−0.11, 0.17]	[−0.04, 0.15]
const	1.14	4.30	7.61
	(3.25)	(2.11)	(1.40)
	[−5.30, 7.58]	[0.11, 8.49]	[4.83, 10.39]
Num.Obs.	115	115	115
R^2	.88	.70	.96
R^2 Adj.	.87	.65	.96
RMSE	0.36	0.24	0.16
Log.Lik.	−46.53	2.89	50.06
Selected lag (p) = 5			
Max root modulus = .96 (stable)			
Portmanteau (serial) $\chi^2(99) = 92.09$, p = .676			

one column per equation (response series); each cell stacks the estimate, its (SE), and the [95% CI] — the per-coefficient t/p are dropped. orthogonalised (Cholesky) IRFs depend on the ordering of the series; a level VAR assumes stationarity. stability check: max companion-eigenvalue modulus < 1 indicates a stable VAR. serial-correlation check: the Portmanteau test (vars::serial.test, asymptotic) tests the null of no residual autocorrelation — a small p suggests remaining serial correlation (consider a higher lag order).

Table 3. Forecast-error variance decomposition

Variable	Impulse	Share
gdp_growth	gdp_growth	.99
gdp_growth	inflation	.00
gdp_growth	unemployment	.01
inflation	gdp_growth	.68
inflation	inflation	.29
inflation	unemployment	.04
unemployment	gdp_growth	.93
unemployment	inflation	.01
unemployment	unemployment	.06

Figure. Dynamic response



Understanding the numbers

Lag. A candidate lag order (how many past periods are included) compared by information criteria to pick the model. Here, lag = 5.

AIC. An information criterion used (with BIC and HQ) to pick the lag order - lower is better relative to the alternatives compared. Here, AIC = −8.30 at this lag.

BIC. Like AIC but penalizes complexity more heavily; used alongside it to pick the lag order. Here, BIC = −7.12 at this lag.

HQ. The Hannan–Quinn criterion: another information criterion for comparing candidate lag orders. Here, HQ = −7.82 at this lag.

Series 1 (equation column). The estimated coefficient for a lagged predictor in an equation - a VAR fits one regression per series, one column per equation. Here, the first equation's estimate for its first term is 0.44.

Num.Obs.. The number of observations used to fit each equation. Here, N = 115.

R^2. How much of an equation's series is explained by the joint lagged predictors (each equation is fit like its own regression). Here, R^2 = .88 for the first equation.

R^2 Adj.. That R^2, adjusted for the number of predictors in the equation. Here, adjusted R^2 = .87 for the first equation.

RMSE. The typical size of an equation's prediction error, in the series' own units. Here, RMSE = 0.36 for the first equation.

Log.Lik.. The log-likelihood for an equation's fit. Here, Log.Lik. = −46.53 for the first equation.

Variable. Which series' forecast-error variance is being decomposed in this row. Here, this row decomposes gdp_growth.

Impulse. Which series' shock is contributing to that variance - the same ordering sensitivity as the impulse-response functions applies here. Here, the shock comes from gdp_growth.

Share. The share of the forecast-error variance attributable to that shock, at the chosen IRF horizon. Here, that share is .99.

Max root modulus. The largest companion-matrix eigenvalue modulus - below 1 means the VAR system is stable. Here, the max root modulus is .96 (stable).

Portmanteau (serial) χ^2. The Portmanteau test statistic for residual serial correlation - a small p suggests remaining autocorrelation the model has not captured. Here, $\chi^2(99) = 92.09$, $p = .676$.

How to read this test

A VAR models several series as functions of their joint past. The impulse-response plots show how a shock to one series propagates to the others over time; read each IRF together with its bootstrap confidence band, and note that orthogonalised (Cholesky) IRFs depend on the ordering of the series. Lag selection picks how many past periods to include. A level VAR assumes stationary series — difference them (or use a VECM) if they are not. The Portmanteau test (vars::serial.test) checks the residuals for remaining serial correlation: a small p suggests the lag order is too low to whiten the residuals. Method: R vars package (Pfaff, 2008; Lütkepohl, New Introduction to Multiple Time Series Analysis).

APA template: A VAR(5) model was selected by AIC; impulse-response functions are shown (Figure).

Statistical basis: Vector autoregression (VAR) estimation (Pfaff B (2008). "VAR, SVAR and SVEC Models: Implementation Within R Package vars." Journal of Statistical Software, 27(4).); VAR modeling framework and impulse-response interpretation (Lütkepohl, H. (2005). New Introduction to Multiple Time Series Analysis. Springer-Verlag, Berlin.). Full references in the exported CITATIONS.txt.